Lecture 13 Nonparametric Regression II

STAT 441: Statistical Methods for Learning and Data Mining

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Outline

1. Kernel Smoothing
2. Local Regression
3. Summary and Remark

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Nonparametric (kernel) density estimation

- A histogram is a nonparametric estimate of the density of y
  - break data into bins of width $2h$
  - form rectangles of area the relative frequency $=$ freq $/n$
  - the height is freq $/2nh$ (check: area $=$ (freq $/2nh) \times 2h =$ freq $/n$).
- Use freq $= \sum_{i=1}^{n} \mathbf{1}\left(x_0 - h < x_i < x_0 + h\right)$
  - where indicator function $1(\mathbf{A})$ equals 1 if event $\mathbf{A}$ happens and equals 0 otherwise
- The histogram estimate of $f\left(x_0\right)$, the density of x evaluated at x_0, is

$$\widehat{f}_{HIST}\left(x_0\right) = \frac{1}{2nh} \sum_{i=1}^{n} \mathbf{1}\left(x_0 - h < x_i < x_0 + h\right)$$

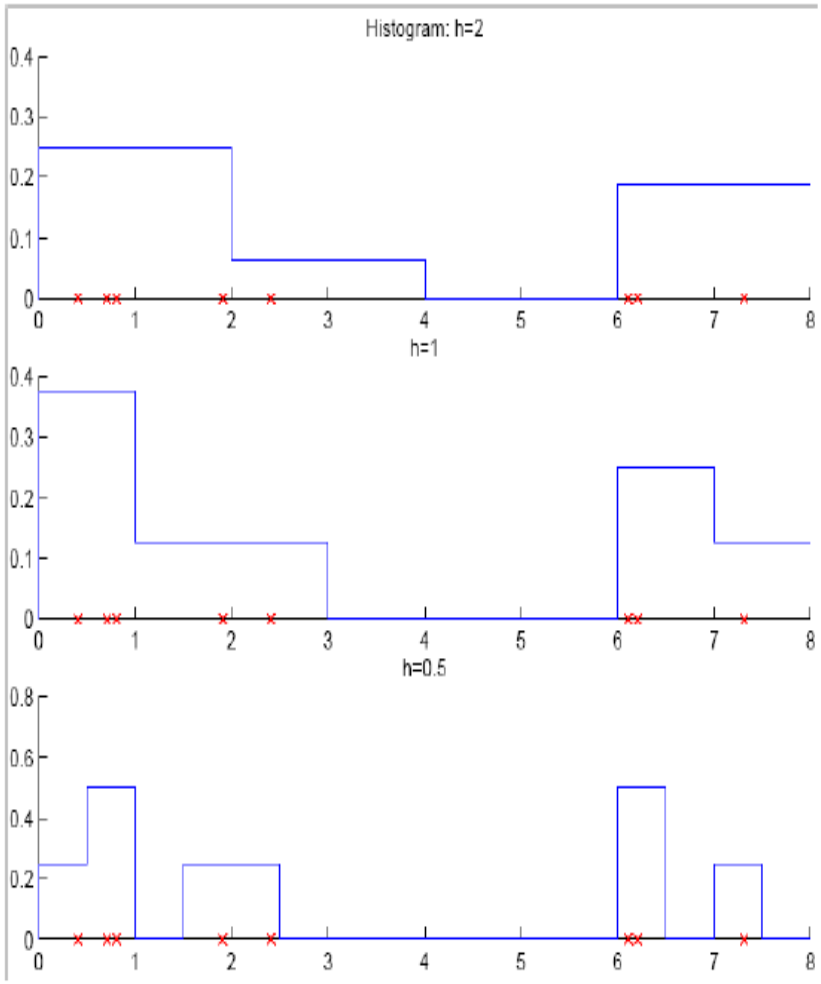
$$= \frac{1}{nh} \sum_{i=1}^{n} \frac{1}{2} \times \mathbf{1}\left(\left|\frac{x_i - x_0}{h}\right| < 1\right).$$

4. The "Boxcar" Kernel

The very last line of math introduces the concept of a **Kernel**. The term $\left|\frac{x_i - x_0}{h}\right| < 1$ is just a mathematical way to define a "box."

- If a point is within distance h of the center x_0, it gets a weight of $1/2$.
- If it's further away, it gets a weight of 0.

Histogram



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Kernel density estimate

- Recall $\widehat{f}_{\text{HIST}}(x_0) = \frac{1}{nh}\sum_{i=1}^{n}\frac{1}{2}\times\mathbf{1}\left(\left|\frac{x_i-x_0}{h}\right|<1\right)$
- Replace $\mathbf{1}(A)$ by a kernel function
- Kernel density estimate of $f(x_0)$, the density of x evaluated at x_0, is

$$\widehat{f}(x_0) = \frac{1}{nh}\sum_{i=1}^{N} K\left(\frac{x_i - x_0}{h}\right)$$

  - $K(\cdot)$ is called a kernel function
  - h is called the bandwidth or window width or smoothing parameter h
- More generally kernel function must satisfy conditions including
  - Continuous, $K(z) = K(-z)$, $\int K(z)dz = 1$, $\int K(z)dz = 1$, tails go to zero.

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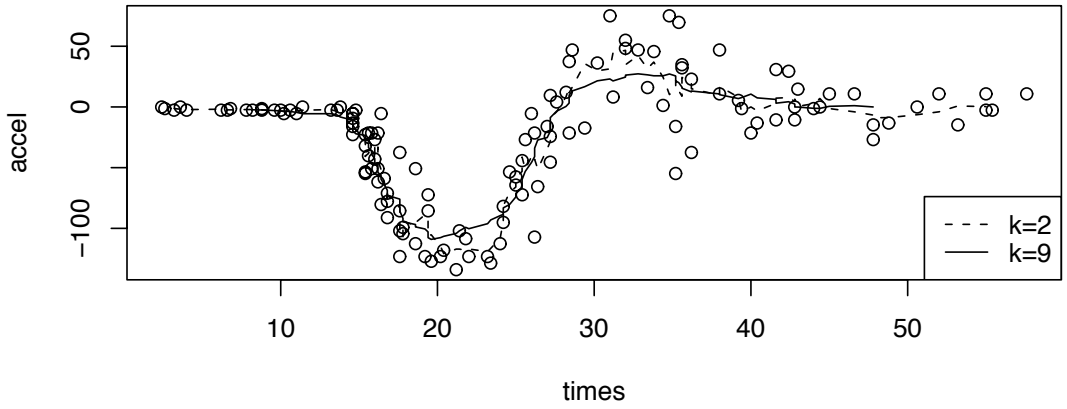
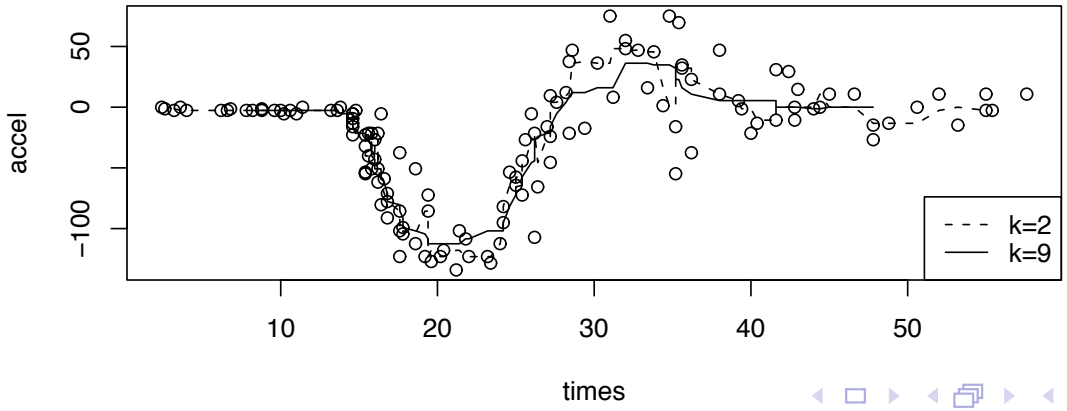
Kernel Smoothing

- When there is no parametric model relating the fitted values at one point to those at other points, it is reasonable to let the fit at x be determined by those points (x_i, y_i) with x_i close to x.
- A first attempt might be running means, in which $\hat{y}_j$ is the average of the y_i with $|i - j| \leq k$ (assuming that $\cdots x_i \leq x_{i+1} \cdots$).
- Alternatively, running medians.
- Then `plot(...,type="l")` for linear interpolation between the $(x_j, \hat{y}_j)$.

Motorcycle Data

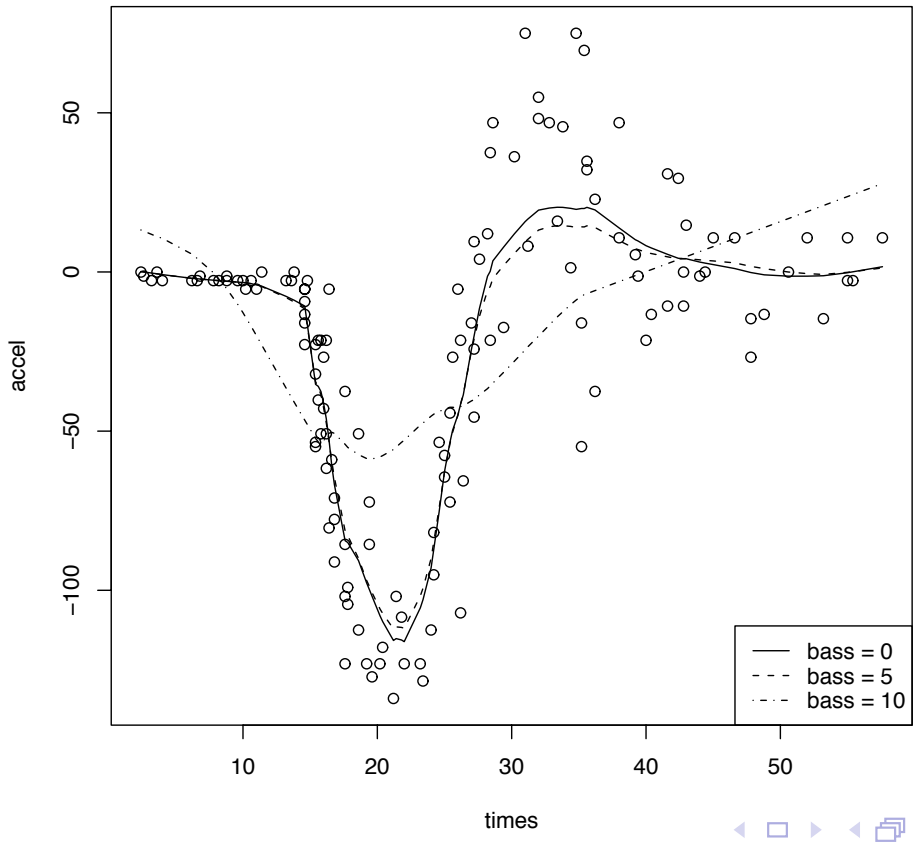
- Running means (top) and medians (bottom)





Motorcycle Data

- The super smoother function `supsmu(...)` on R will replace running means with running linear regressions - at each point (x_i, y_i), $\hat{y}_i$ is obtained by doing a linear regression using only k nearby points as data.



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Kernel Smoothing

- More flexible is kernel smoothing, in which the fitted value at x is a weighted average of those values of y observed at points x_j near x:

$$\hat{y}(x) = \sum_{i=1}^{n} w\left(x - x_i\right) y_i,$$

- where $w(x - x_i)$ is typically a symmetric function
  - $\sum_{i=1}^{n} w\left(x - x_i\right) = 1$.
  - decrease as the distance between x_i and x_0 increases
  - place more weight on observations with x_i close to x_0 as bandwidth h decreases
- The Nadaraya-Watson kernel uses

$$w(x - x_i) = \frac{K_h\left(x - x_i\right)}{\sum_{i=1}^{n} K_h\left(x - x_i\right)},$$

where $K(t)$ is a unimodal probability density, symmetric about 0, and $K_h(t) = \frac{1}{h} K\left(\frac{t}{h}\right)$.

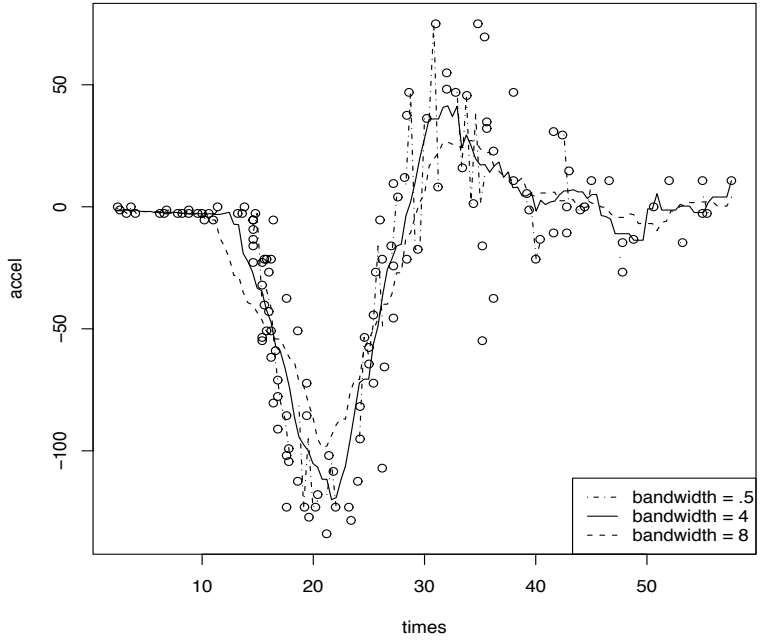
Kernel Functions

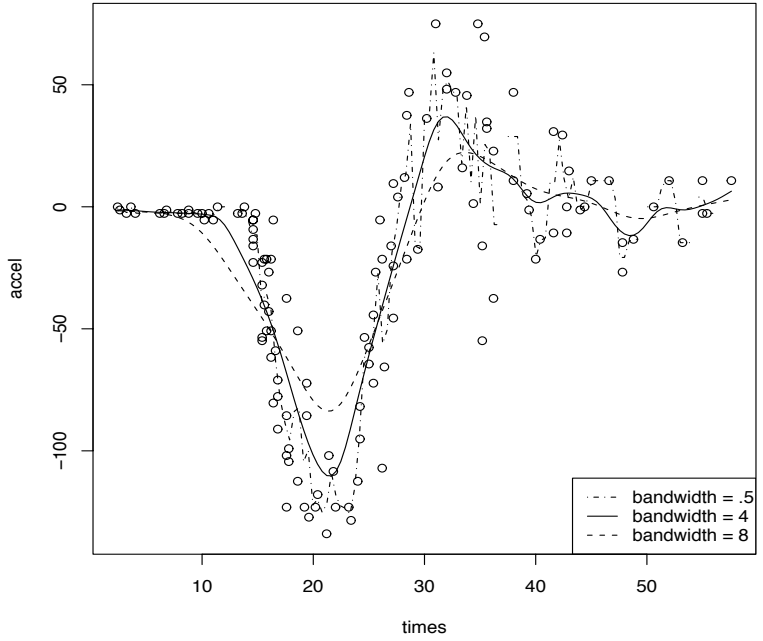
- Common choices of kernel functions:
  1. Epanechnikov kernel: $K(t) = \frac{3}{4}\left(1 - t^2\right) I(|t| \leq 1)$.
  2. Tri-cube function: $K(t) \propto \left(1 - |t|^3\right)^3 I(|t| \leq 1)$.
  3. Uniform (box in R): $K(t) = .5I(|t| \leq 1)$. - running mean?
  4. Gaussian: $K(t) = \phi(t)$.
- In R one can choose a bandwidth, which is a monotonic function of λ defined by

$$.75 = \int_{-\infty}^{\text{bandwidth/4}} K_\lambda(x)dx.$$

Motorcycle Data

- Kernel smooths to motorcycle data; box kernel (left) and normal kernel (right). Bandwidth = .5 is the default.





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Kernel Smoothing

- Kernel smooths can be badly biased near the edges of the region containing the xs (since there are too few x_is on one side of x).
- Without special conditions on the design (the choice of the x_i) or on the kernel, they can be badly biased elsewhere.
- In recent years attention seems to have shifted away from kernel smoothing and towards local regression methods.

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Local Regression

- A sample mean of y = OLS of y on an intercept.
- A weighted sample mean of y = weighted OLS of y on an intercept.
- Suppose we have data $(x_i, y_i = f(x_i) + \varepsilon_i)$.
- Consider estimating $f(x_0)$ by a constant $\hat{\theta}(x_0)$ (the kernel (local constant) estimator) defined by

$$\hat{\theta}(x_0) = \arg\min_{\theta} \sum_{i=1}^{n} K_h(x_0 - x_i)(y_i - \theta)^2 .$$

- Then

$$\hat{\theta}(x_0) = \sum_{i=1}^{n} \frac{K_h(x_0 - x_i)}{\sum_{i=1}^{n} K_h(x_0 - x_i)} y_i,$$

  the kernel smoother.
- This is then a special case of local regression - Locally Constant.

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Local Regression

- The local linear estimator generalizes to local constant estimator
- A locally linear fit is

$$\begin{aligned}\hat{f}(x_0) &= \hat{\theta}_0(x_0) + \hat{\theta}_1(x_0)\, x_0 \text{ for} \\ \hat{\boldsymbol{\theta}}(x_0) &= \arg\min_{\theta} \sum_{i=1}^{n} K_h(x_0 - x_i)\left(y_i - \theta_0 - \theta_1 x_i\right)^2;\end{aligned}$$

  a locally quadratic fit includes $\theta_2 x_i^2$, etc.
- For general multiple regression with regressors $\mathbf{x}$ one solves

$$\hat{\boldsymbol{\theta}}(\mathbf{x}_0) = \arg\min_{\theta} \sum_{i=1}^{n} K_h(\mathbf{x}_0, \mathbf{x}_i)\left(y_i - \left(1, \mathbf{x}_i^T\right)\theta\right)^2$$

  and sets $\hat{f}(\mathbf{x}_0) = \left(1, \mathbf{x}_0^T\right)\hat{\boldsymbol{\theta}}(\mathbf{x}_0)$;
- $K_h(\mathbf{x}_0, \mathbf{x}_i)$ is typically radially symmetric, i.e. a function of $\|\mathbf{x}_0 - \mathbf{x}_i\|$ such as $\frac{1}{h}\phi\left(\frac{\|\mathbf{x}_0 - \mathbf{x}_i\|}{h}\right)$.

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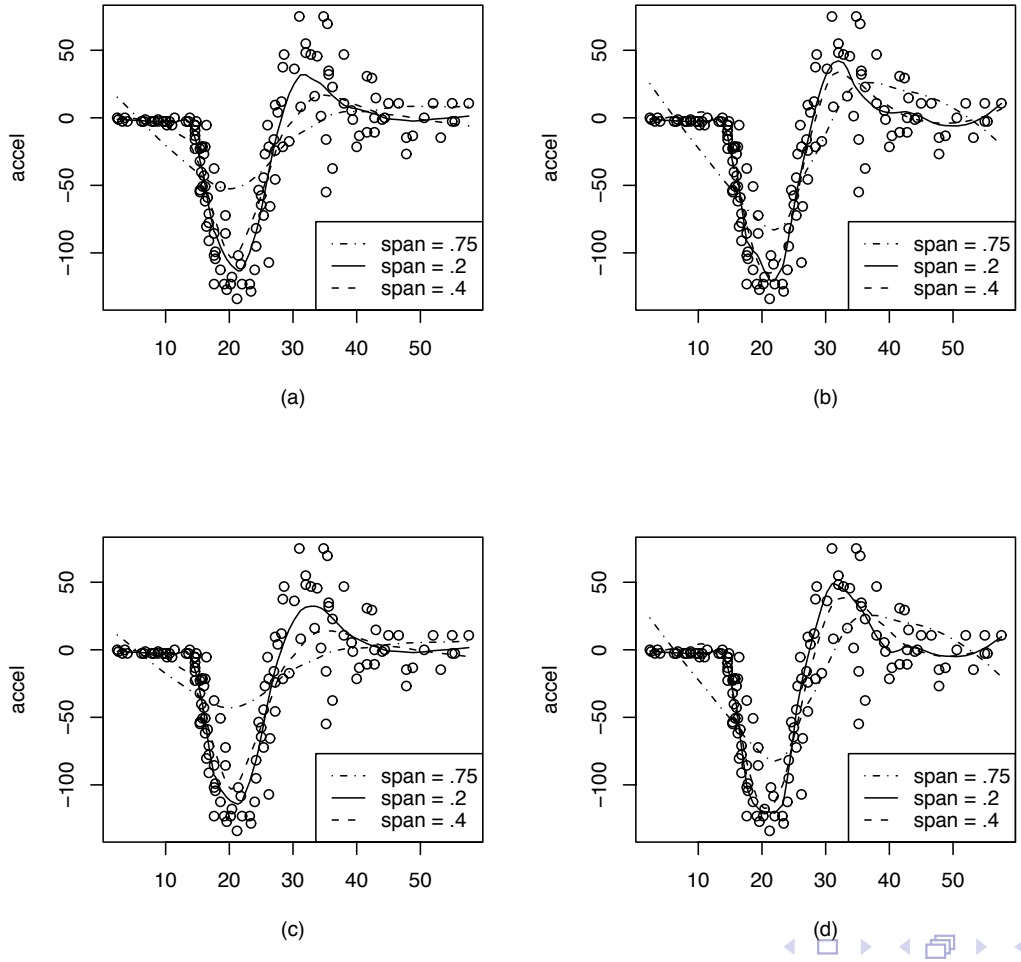
Local Regression

- An advantage of local regression estimators over kernel smoothing is in bias reduction.
- The variance ($\text{VAR}[\hat{f}(x_0)]$) increases as more terms are added; there is a trade-off between bias and variance.
- We have $\hat{\mathbf{y}} = \mathbf{S}_\lambda \mathbf{y}$, where $\mathbf{S}_\lambda$ has rows $\mathbf{b}^T(x_i)$; as for splines λ can be chosen by cross-validation (but isn't on R).
- A (possibly) robust version of local polynomial regression (for $r = 0, 1, 2$) is incorporated in R, as the function `loess(...)`. Important options are

Loess fits

- Locally linear; gaussian family (a) and symmetric family (c). Locally quadratic; gaussian family (b) and symmetric family (d).



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Summary and Remark

- Kernel Smoothing
- Local Regression
- Read textbook Chapter 7 and `R` code